Project Overview

City governments receive thousands of service requests from residents every day. These requests may involve noise complaints, sanitation issues, building problems, street conditions, public health concerns, and many other types of local service needs. Because city agencies have limited time and resources, one important analytics challenge is identifying which requests are likely to take longer to resolve.

In this project, you will work with a real-world sample of NYC 311 service request data. The dataset contains service requests submitted during April 2025 and includes information such as when each request was created, when it was closed, the responding agency, the type of problem reported, the location category, the borough, and the channel through which the request was submitted.

Your task is to prepare the data for analysis, explore the structure and quality of the dataset, develop predictive models, and translate your results into a practical business recommendation for city operations leaders.

Because this is the first data mining project in the course, you will also complete a project worksheet. The worksheet is designed to guide your team through the major steps of the project and help you document the decisions you make along the way. You should complete the worksheet as you work through the project, not after everything is finished.

Required Analysis

Your analysis should include the following major steps.

1. Business and Data Understanding

Begin by making sure you understand the business problem and the dataset. You may need to do a small amount of background research to understand what NYC 311 service requests are, how residents use the system, and why resolution time might matter to city operations leaders.

You should also review the data dictionary and identify what each major variable represents. At this stage, pay attention to which variables would be known when a service request is first submitted and which variables would only be known later in the process.

The project worksheet will guide you through this step by asking you to explain the business problem, define the prediction goals, and identify how city leaders might use the results.

2. Exploratory Data Analysis

Before building models, you need to explore the dataset carefully. This step should take a meaningful amount of your project time because good models depend on good understanding of the data.

Your exploratory data analysis should help you answer questions such as:

- How many rows and columns are in the dataset?
- Which variables have missing values?
- Are any variables mostly missing or not useful for modeling?
- Which categorical variables have many categories or rare values?
- What are the most common agencies, problem types, boroughs, location types, and submission channels?
- Are service requests evenly distributed across the month, days of the week, or hours of the day?
- What does the distribution of resolution time look like?
- Are there very short or very long resolution times?
- Does resolution time differ by agency, problem type, borough, or submission channel?
- Are some variables strongly related to each other, such as agency and problem type?

You should create tables and visualizations to support your answers. At minimum, your exploratory work should include missing value summaries, frequency tables or bar charts for key categorical variables, histograms of resolution time, comparisons of resolution time across important groups, and at least two cross-tabulations between important categorical variables.

The goal of this step is not just to make charts. The goal is to learn what the data are telling you before you model. Your EDA should help you decide which records to remove, which variables to keep, which variables to simplify, whether resolution time should be transformed, and how to define a “slow” service request.

The project worksheet will guide you through the required EDA items and ask you to explain what you learned from them.

3. Data Preparation

After exploring the data, prepare the dataset for modeling. At minimum, you should:

- Convert date/time fields into usable formats.
- Create a resolution time variable using the created and closed dates.
- Remove or address records that cannot be used for modeling.
- Create useful date/time features, such as day of week, hour of day, or weekend indicator.
- Decide how to handle missing values.
- Decide how to handle categorical variables with many categories.
- Identify and remove variables that would create data leakage.

Data leakage occurs when a model uses information that would not actually be available at the time of prediction. For this project, assume the model would be used when a request is first submitted. Therefore, your predictors should be limited to information that would reasonably be known at that time.

Your data preparation decisions should be documented in the project worksheet and reflected in your analysis script.

4. Linear Regression Model

Build a linear regression model to predict request resolution time. You may choose whether to model raw resolution time or a transformed version of resolution time, but you should justify your choice.

Your evaluation should focus on prediction performance and practical usefulness, not statistical significance alone. You should use appropriate error metrics and explain what the model's performance means in the context of service request management.

5. Logistic Regression Model

Build a logistic regression model to classify whether a request is likely to be slow.

For this project, a "slow" request should be defined in a way that makes business sense. For example, you may define slow requests as those in the slowest 25% of resolution times within their problem category. This approach recognizes that different types of service requests may naturally have different expected resolution times.

You should evaluate the logistic regression model using a confusion matrix and classification performance measures. Be sure to discuss which errors are most important from a business perspective. For example, city leaders may be especially concerned about requests that are predicted to be normal but actually become slow.

6. Business Recommendation

Your final task is to translate your modeling results into a recommendation for city operations leaders.

Your recommendation should address questions such as:

- What types of requests appear most likely to experience longer resolution times?
- How useful is the model for identifying potentially slow requests?
- What tradeoffs would city leaders face if they used this model?
- How could the model support triage, monitoring, or follow-up?
- What limitations should decision-makers understand before relying on the model?

Your recommendation should be grounded in your analysis. Do not overstate what the model can do.

Final Deliverables

Your team will submit three main deliverables.

1. Project Worksheet

- Complete the project worksheet as you work through the analysis.
- The worksheet will guide you through the business problem, data exploration, data preparation, model building, model evaluation, and business recommendation.
- For each section, explain the decisions your team made and why those decisions were appropriate.
- Your responses should be clear enough that someone could understand your process without reading your full code.

2. Analysis Script

- Your script should show your complete data preparation and modeling process.
- Your script should include the code needed to produce the tables, visualizations, models, and performance metrics used in your worksheet and presentation.
- Use or adapt the R scripts and methods taught in the course lectures.
- The script should be organized, readable, and reproducible.
- Include comments explaining major steps and decisions.

3. Business Presentation

- Your presentation should communicate the business problem, modeling approach, key findings, model performance, and recommendation.
- The audience is a group of city operations leaders, not data scientists.
- Focus on clear explanation, practical interpretation, and responsible use of the model.
- Your presentation should be consistent with the analysis shown in your worksheet and script.

Grading Rubric

Total: 100 points

Component	Points	What I am looking for
Project Worksheet	35	Complete, thoughtful responses that explain your data exploration, preparation decisions, leakage check, modeling choices, evaluation, and business interpretation.
Analysis Script	25	A clear, organized, reproducible script that imports the data, explores the data, prepares variables, builds both required models, and evaluates model performance.
Business Presentation	30	A clear presentation for a business audience that explains the problem, key data findings, model results, recommendation, limitations, and responsible use of the model.
Consistency and Professionalism	10	The worksheet, script, and presentation are consistent with one another, complete, well organized, and submitted in the requested format.

Important Expectations

Your project should show that you can:

- Calculate and explain resolution time.
- Explore missing values, categorical variables, date/time patterns, and resolution time distributions.
- Identify and avoid data leakage.
- Prepare a clean modeling dataset.
- Build and evaluate both a linear regression model and a logistic regression model.
- Interpret model results from a data mining and business decision-making perspective.
- Make a recommendation that is supported by your analysis.

The project is not graded on finding a perfect model. It is graded on the quality of your process, the reasonableness of your decisions, and your ability to explain how the analysis could support a real operational decision.